Prediction of Disulfide Connectivity From Protein Sequences

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Abstract (1/2)

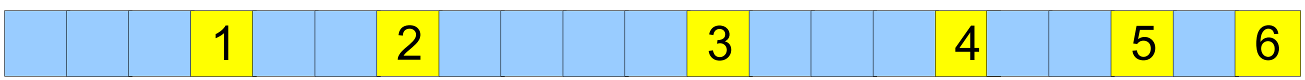
- The difficulties in predicting disulfide connectivity from protein sequences lie in the nonlocal properties of the disulfide bridges that involve cysteine pairs at large sequence separation. Though some progress has been recently made in the prediction of disulfide connectivity, the current methods predict less than half of the disulfide patterns for the data set sharing less than 30% sequence identity.

Abstract (2/2)

- In this report, we use the support vector machines based on sequence features such as the coupling between the local sequence environments of cysteine pair, the cysteines sequence separations, and the global sequence descriptor, such as amino acid content. Our approach is able to predict 55% of the disulfide patterns of proteins with two to five disulfide bridges, which is 11–26% higher than other methods in the literature.

Introduction (1/2)

- The disulfide bonds impose geometrical constraints on the protein backbones.
  - For example:



[1-6,2-3,4-5] [1-6,2-4,3-5] [1-5,2-4,3-6]

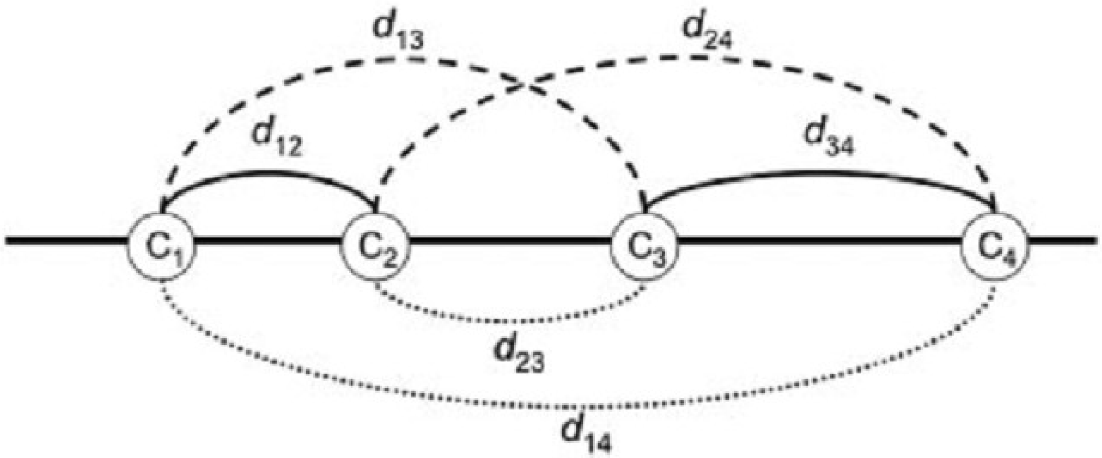
Introduction (2/2)

- The problem is further complicated by the rapid increase of possible disulfide patterns as the number of disulfide bridges increases.
  - B=2 : 3 possible disulfide patterns

$[c_1\text{-}c_2, c_3\text{-}c_4]$

$[c_1\text{-}c_3, c_2\text{-}c_4]$

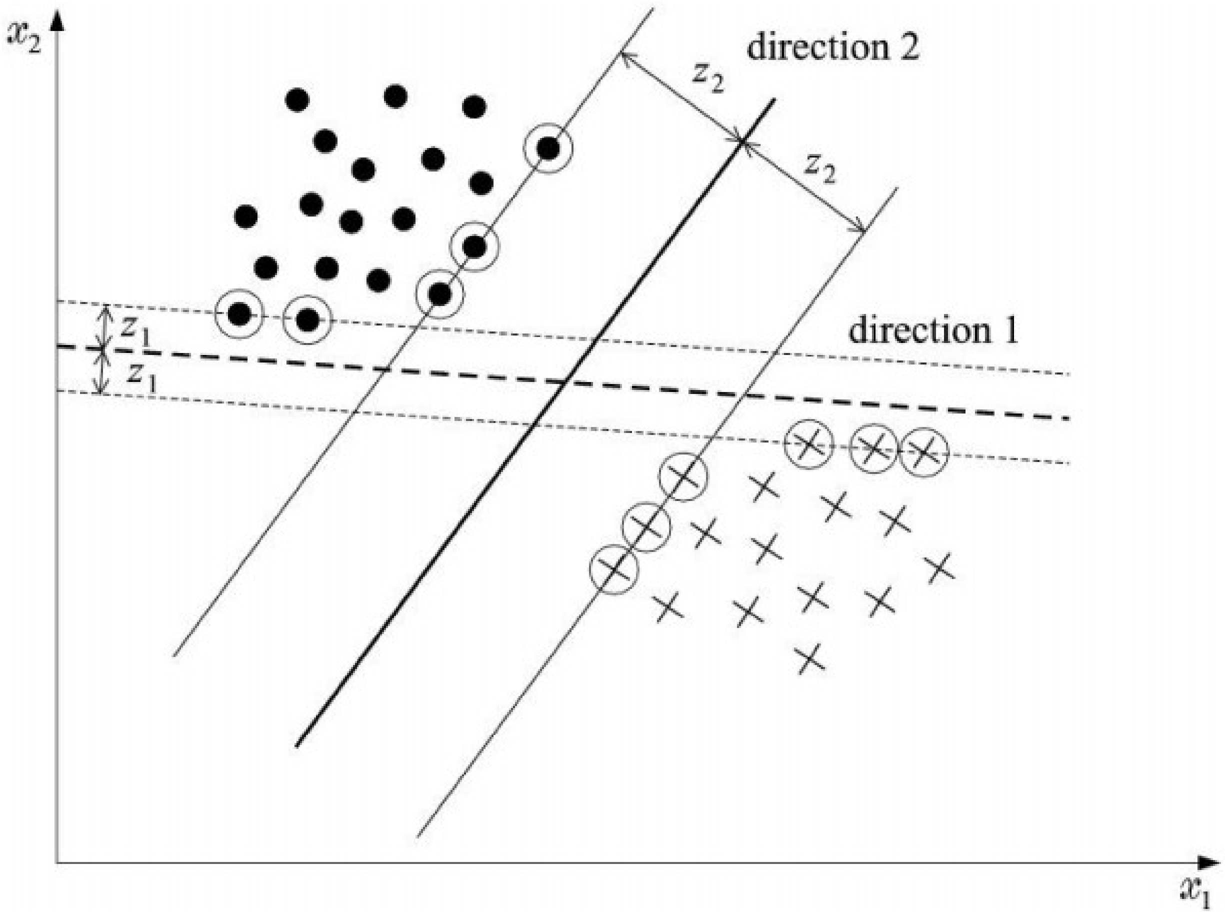
$[c_1\text{-}c_4, c_2\text{-}c_3]$



  - B=3 : 15, B=4 : 105, B=5 : 945

SVM

- The maximum margin interpretation.
- It is a typical constrained optimization problem.

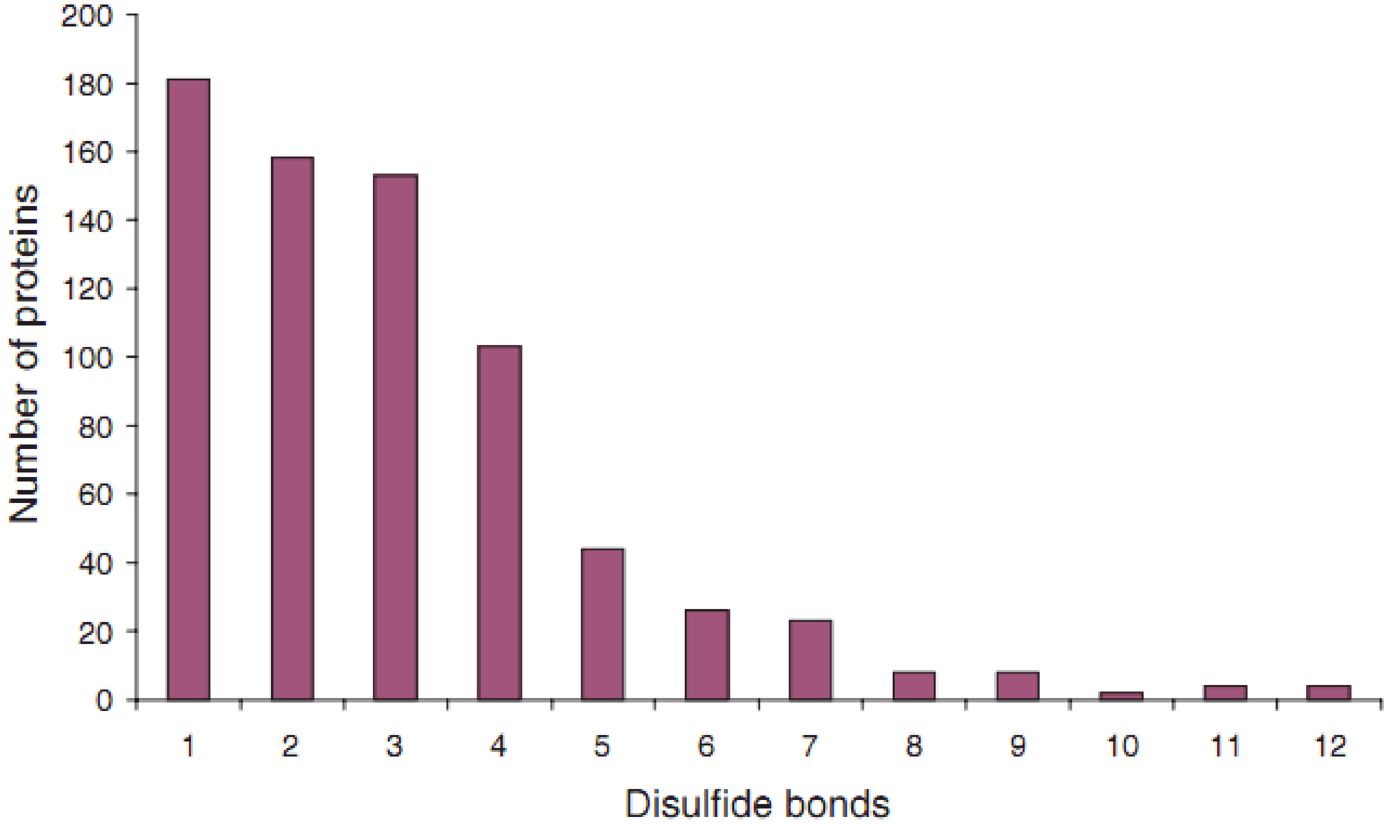


Data Sets (1/3)

- SWISS-PROT databases releases No. 39.
  - 86593 entries from PDB and other databases
- The constructed set contain the sequences
  - only with experimentally verified intra chain disulfide bridge annotations and
  - excludes whose disulfide bonds are assigned as “probable,” “potential,” or “by similarity”

Data Sets (2/3)

- 482 sequences
  - <u>B=2 : 168</u>, <u>B=3 : 177</u>, <u>B=4 : 95</u>, <u>B=5 : 42</u>



Data Sets (3/3)

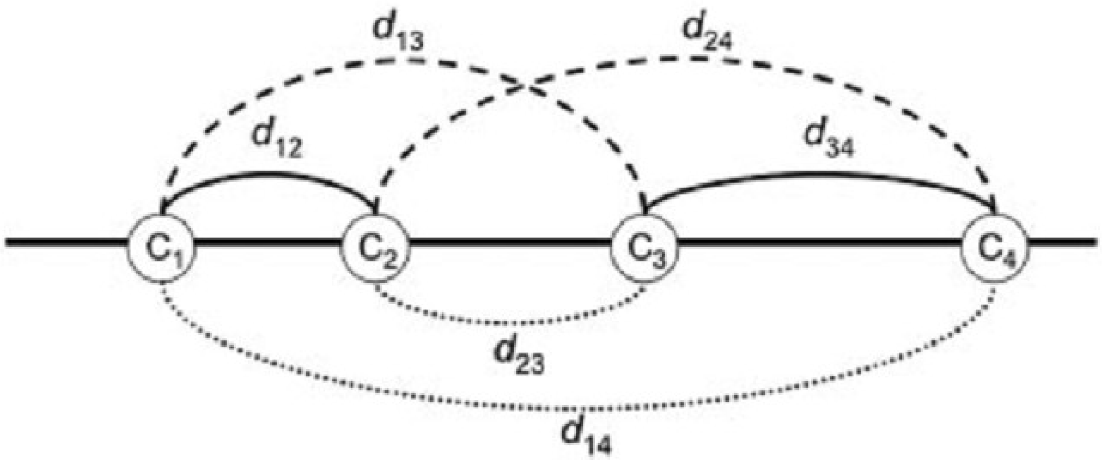
- For the four-fold cross-validation, group sequences into four sets
  - The sequence homology among the sets is less than 30%
  - The number of the sequences of the set is approximately equal

Future Vectors (1/2)

- Cysteine-cysteine coupling (S)
  - A sequence window centered on the cysteine is used to describe the neighboring sequence environment.
  - Evolution information of the protein sequence is using the sequence profile generated by PSSM
    - A 20-element vector
  - Window size setting by preliminary experiment:
    - B=2 : 7
    - B=3, 5 : 21
    - B=4 : 27

Future Vectors (2/2)

- Cysteine spacing patterns (D)



- Amino acid content (A)
  - A 20 composition vector: the probability for each amino acid

$$A = (a_1, a_2, \ldots, a_{20}), \text{ where } a_k = n_k / n_0.$$

Performance Assessment

$$Q_c = \frac{\text{number of correctly predicted bonds}}{\text{number of total bonds}}$$

$$Q_p = \frac{\text{number of correctly predicted proteins}}{\text{number of total proteins}}$$

- Random predictor
  - $Q_c = 1/(2B-1)$
  - $Q_p = 1/Np$, where $N_p = (2B-1)!! = \Pi_{(i \leqslant B)}(2i-1)$

Result (1/3)

- SVM with Single-Feature

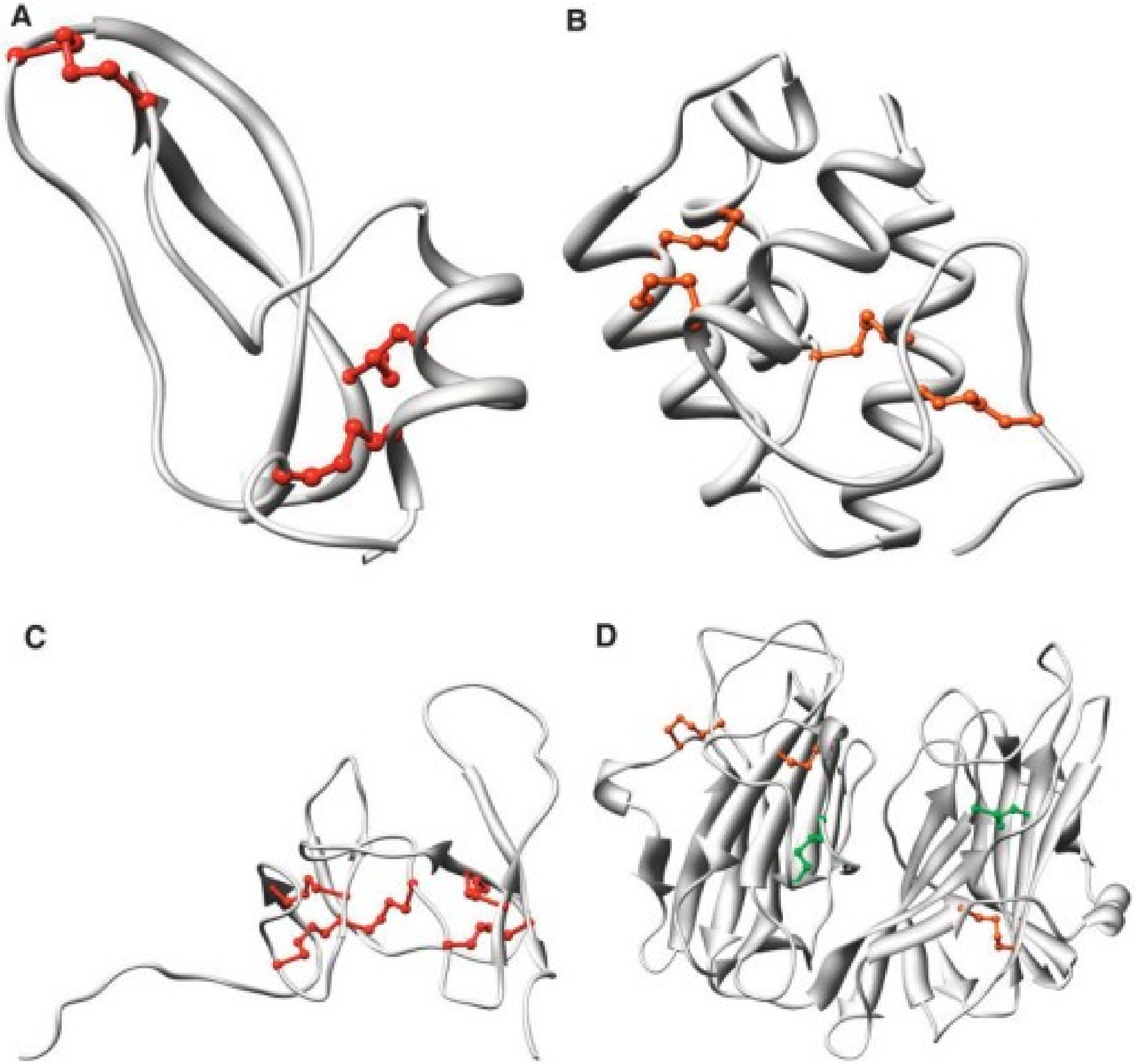
	$B = 2$		$B = 3$		$B = 4$		$B = 5$		$B = 2...5$	
Method	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c
R	0.33	0.33	0.06	0.20	0.01	0.14	0.001	0.11	0.14	0.20
A	0.61	0.61	0.38	0.51	0.13	0.20	0.07	0.27	0.39	0.42
S	0.67	0.67	0.47	0.60	0.17	0.24	0.12	0.32	0.45	0.48
D	0.67	0.67	0.54	0.64	0.28	0.39	0.12	0.30	0.50	0.54

- SVM with Multi-Feature

	$B = 2$		$B = 3$		$B = 4$		$B = 5$		$B = 2...5$	
Method	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c
$D + A$	0.74	0.74	0.54	0.64	0.28	0.39	0.12	0.30	0.52	0.55
$D + S$	0.71	0.71	0.60	0.66	0.30	0.41	0.12	0.30	0.54	0.55
$D + S + A$	0.74	0.74	0.61	0.69	0.30	0.40	0.12	0.31	0.55	0.57

Weight of A = 1, Weight of S = 0.001

Result (2/3)



Result (3/3)

- Comparison with other methods

	$B = 2$		$B = 3$		$B = 4$		$B = 5$		$B = 2...5$	
Method	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c	Q_p	Q_c
MCGM[27]	0.56	0.56	0.21	0.36	0.17	0.37	0.02	0.21	0.29	0.38
NNGM[28]	0.68	0.68	0.22	0.37	0.20	0.37	0.02	0.26	0.34	0.42
RNN[29]	0.73	0.73	0.41	0.51	0.24	0.37	0.13	0.30	0.44	0.49
This work	0.74	0.74	0.61	0.69	0.30	0.40	0.12	0.31	0.55	0.57